

Technical Specification: AI-Enhanced Academic Review System (SCORMETRY 2.0)

1. System Overview

A web-based platform facilitating structured paper submissions, multi-reviewer grading via AI-extracted rubrics, and conditional public visibility based on performance thresholds and teacher approval.

2. Core Entities & Database Schema (Relational)

Users

- `id, name, email, role` (Admin, Teacher/Judge, Reviewer, Student)

Subjects (Rooms)

- `id, title, description, teacher_id`
- `passing_score` (Integer: threshold for public visibility)

Subject_Members

- `subject_id, user_id, role_in_subject` (Reviewer or Student)

Rubrics (Digital Rubric)

- `id, subject_id`
- `pdf_path` (Original official file)
- `structure_json` (Extracted schema: `criteria, levels, weights`)
- `status` (Enum: `uploaded, pending_verification, locked`)

Papers & Reviews

- `papers: id, student_id, subject_id, final_score, visibility_status`
- `reviews: id, paper_id, reviewer_id, scores_json, is_submitted`

3. Feature Breakdown: AI Rubric Workflow (Referencing Diagram)

F1: AI Extraction & Verification (Judge/Teacher)

As per the **SCORMETRY 2.0 Workflow**, the process follows these sequential steps:

- Upload:** Judge uploads the "Official Rubric PDF" to the Room.
- AI Extraction (Gemini):** - System sends PDF to Gemini with an extraction prompt.
 - Gemini parses the document into a **Structured JSON Rubric**.
- Validate Structure:** Backend validates the JSON against a required schema (ensuring total weights = 100% and max points are integers).

4. **Save as Digital Rubric:** The data is saved in a "Pending" state.
5. **Verify and Approve:** The Judge/Teacher reviews the digital table, makes manual corrections if necessary, and clicks **Approve**.
6. **Lock Rubric:** Once approved, the rubric is locked. No further edits are allowed to ensure grading integrity for that subject.

F2: Student & Reviewer Visibility

- **Students:** Can access the "Official Rubric PDF" for guidance but cannot see the "Digital Rubric" (the underlying scoring weights/JSON structure) to prevent "gaming" the system.
- **Reviewers:** Use the "Digital Rubric" as a dynamic interface for scoring submitted papers.

F3: Final Score Calculation

- **Trigger:** When all assigned reviewers submit their `scores_json`.
- **Logic:** `Final_Score = Average(Reviewer_Total_Scores)`.
- **Public Release:** If `Final_Score >= passing_score`, the Teacher is prompted to "Confirm Publication" to make the paper public.

4. API & Integration Layer

Gemini AI Integration

- **Model:** `gemini-2.5-flash-preview-09-2025`
- **Task:** Vision/Text analysis of rubric PDFs.
- **Payload:** System instructions to output only JSON containing `criteria`, `description`, `max_score`, and `level_descriptors`.

File Storage

- Secure bucket storage for PDFs.
- Use of signed URLs for restricted access to student papers under review.

5. Implementation Roadmap

1. **Sprint 1:** Auth & Subject/User Management.
2. **Sprint 2:** PDF Upload & **Gemini Extraction Pipeline** (as per diagram).
3. **Sprint 3:** Digital Rubric Verification UI & Locking Logic.
4. **Sprint 4:** Student Submission & Final Score Calculation Engine.